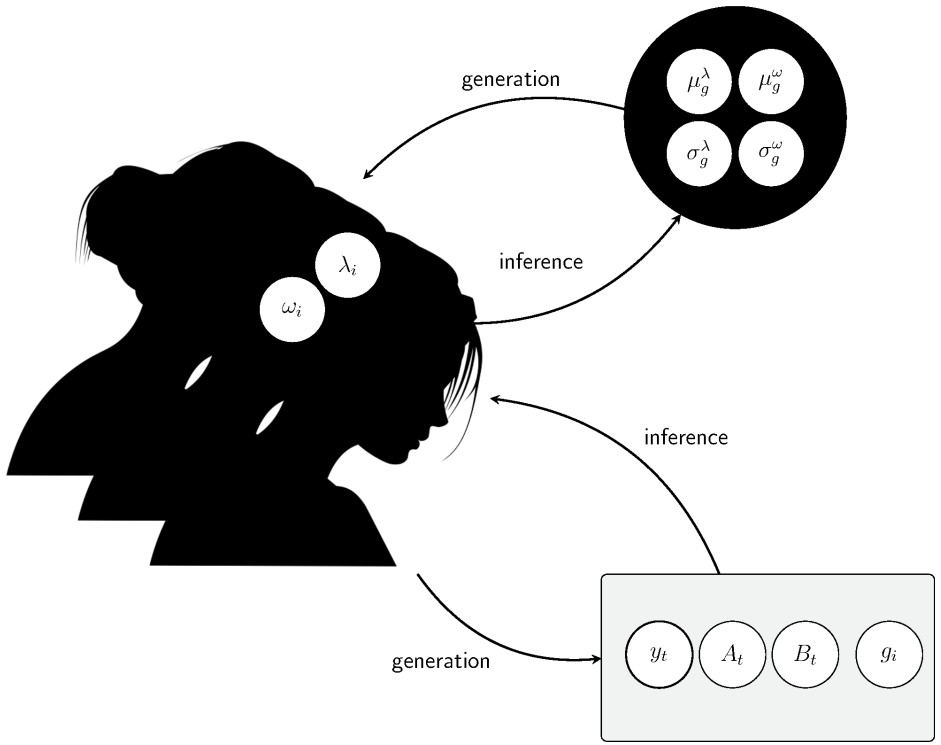
The Hierarchical Logistic Choice Model

Michael D. Lee

Hierarchical Logistic Choice Model Structure

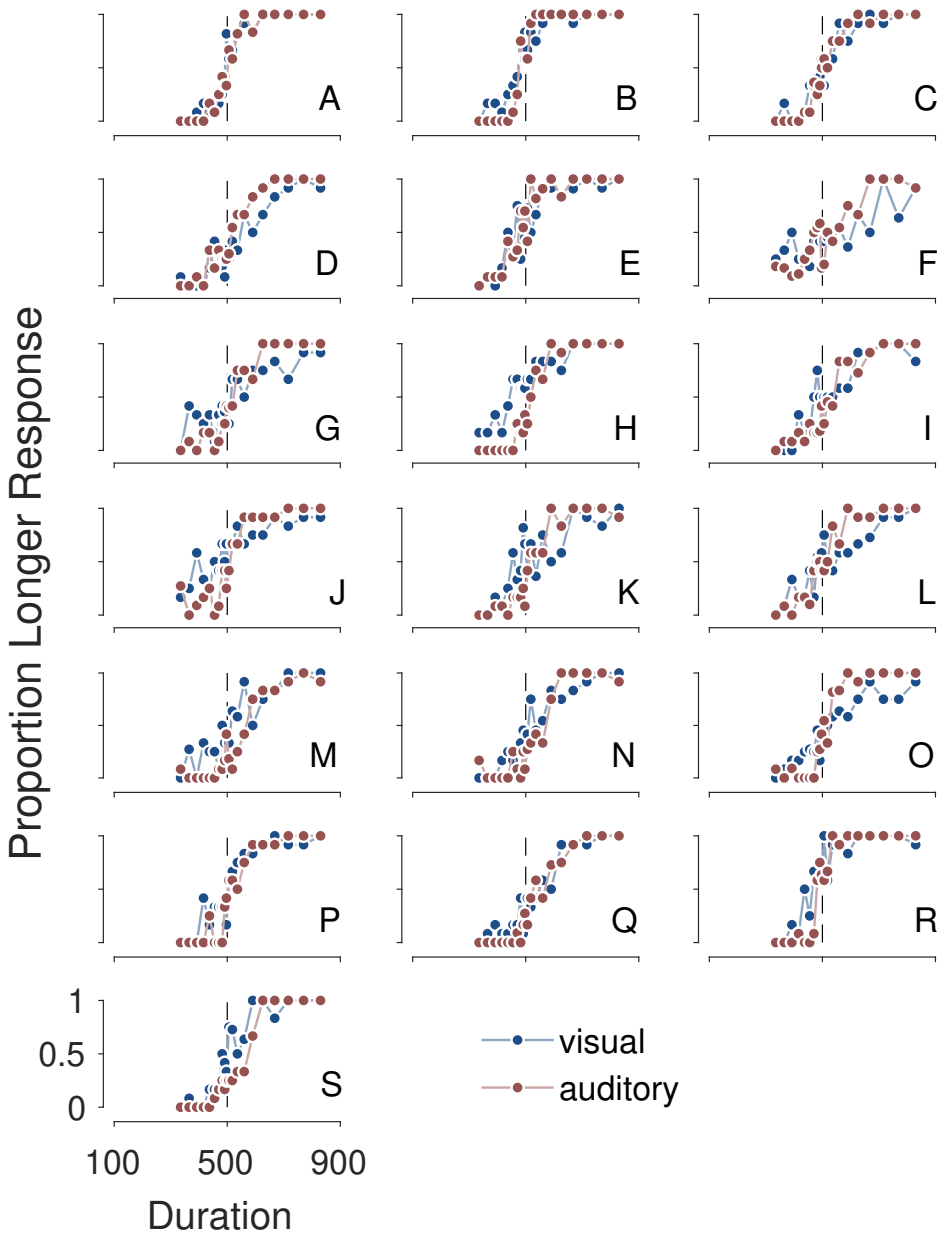
- The hierarchical logistic choice model has the structure
  - observed binary decisions y_{it} on trial t for individual i, for alternatives with evidence A_{it} and B_{it}
  - known allocation of each individual to a group g_i
  - individual weight ω_i and consistency λ_i parameters
  - the weight and consistency parameters come from Gaussian group distributions with means μ_g^ω, μ_g^λ and standard deviations σ_g^ω, σ_g^λ



Modality Duration Perception

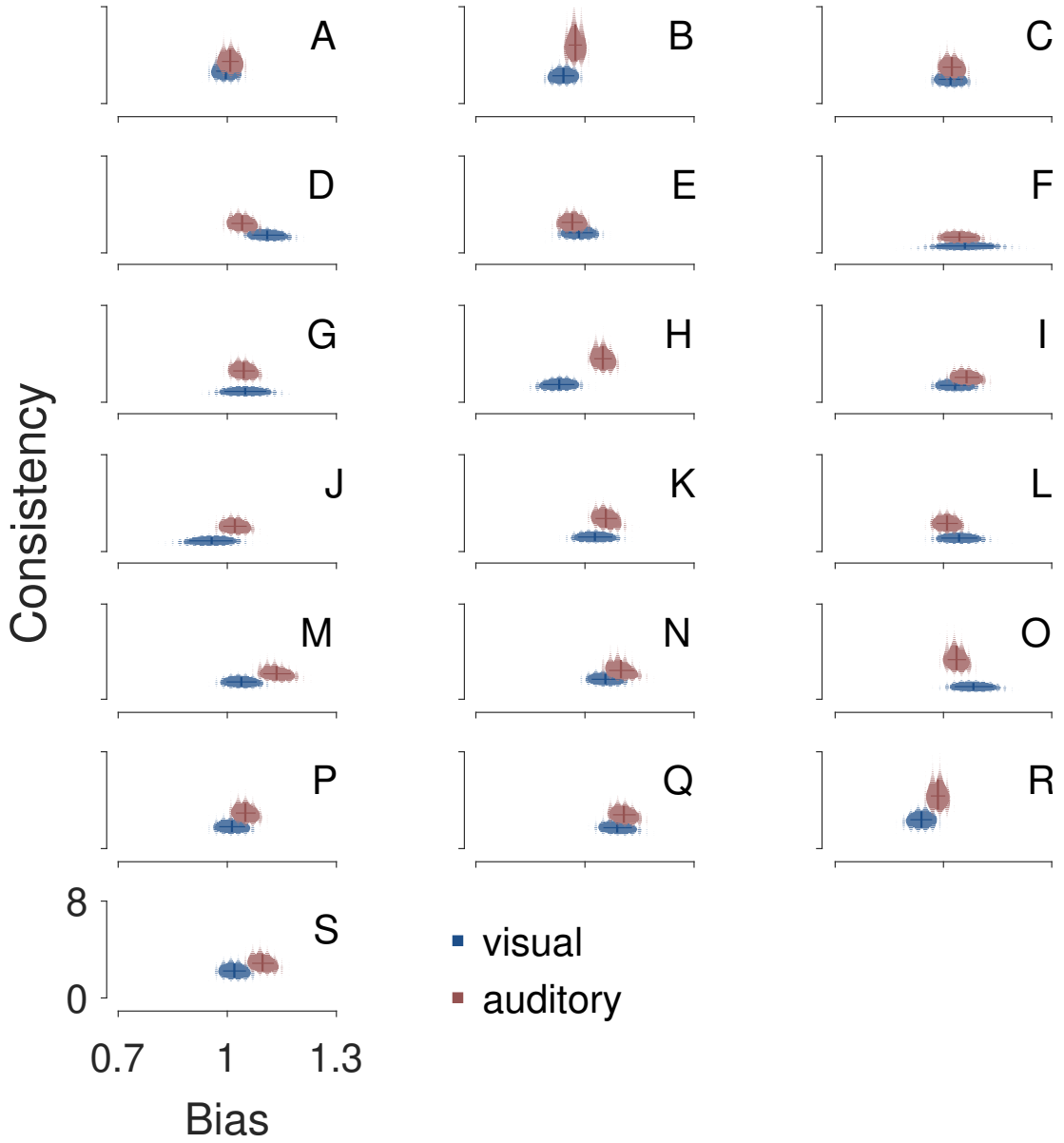
Duration Perception Data

- van Driel et al. (2014) studied duration perception
  - 240 trials in auditory and visual modalities
  - 20 stimuli each presented 12 times that varied in duration from 335ms to 830ms
  - always compared to the 500ms standard
- Behavior of all 19 participants is shown for both modalities
  - lots of individual variability, and hard to see systematic differences in consistency and bias



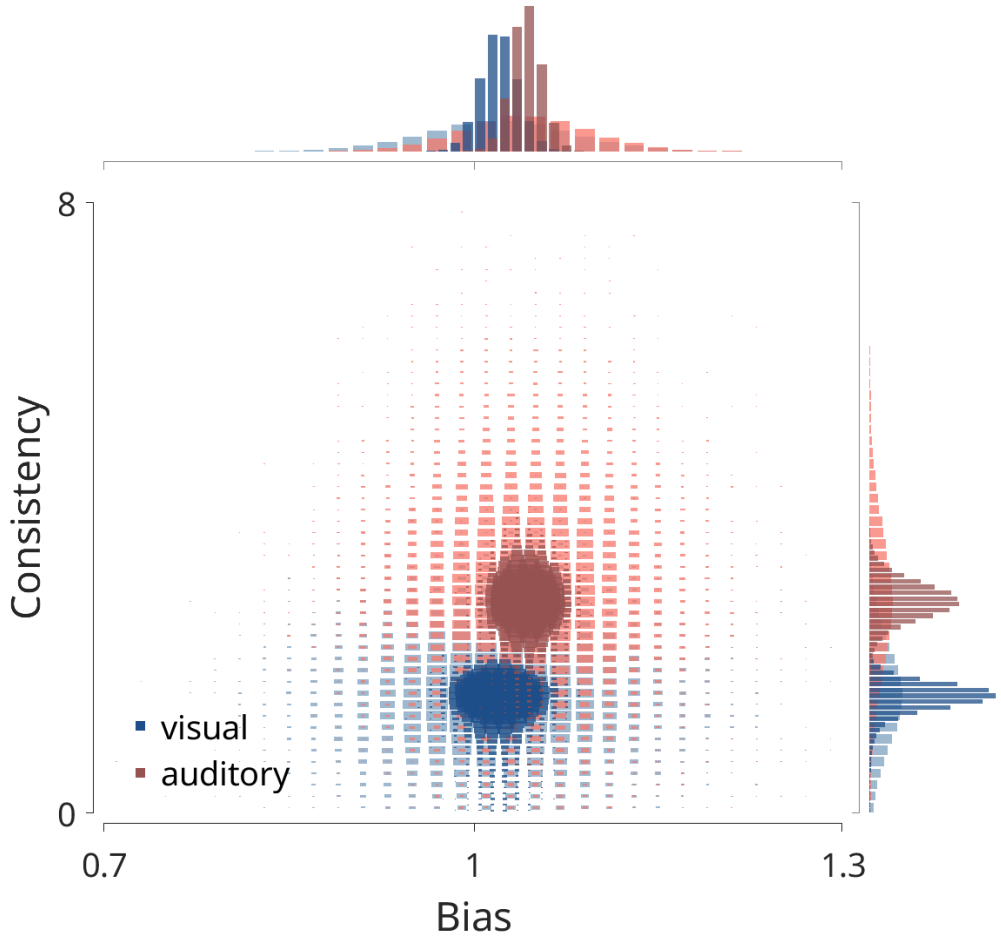
Individual Consistency and Bias

- For most individual participants, there is evidence their duration perception varies across modalities
  - often more consistent in making auditory discriminations
  - sometimes biased towards "shorter" response in auditory discrimination



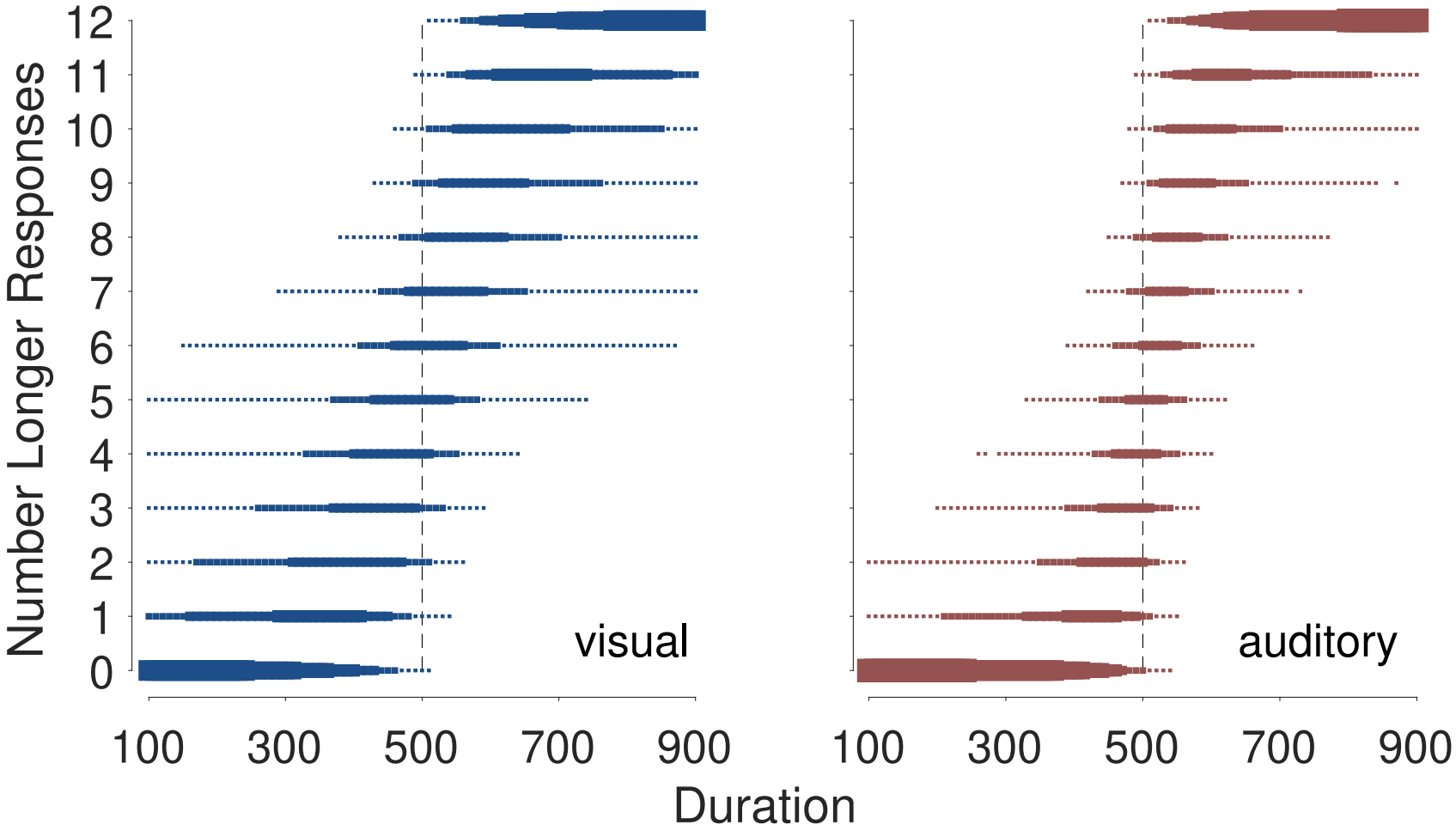
Group and Representative Consistency and Bias

- The group inferences show a likely difference in the mean consistency between durations
  - useful scientific insight that is not obvious from simple behavioral analysis



Representative Individual Posterior Predictions

- The representative inferences emphasize individual differences
  - limited ability to reach strong conclusions at the individual level



References i

van Driel, J., Knapen, T., van Es, D. M., & Cohen, M. X. (2014). Interregional alpha-band synchrony supports temporal cross-modal integration. *NeuroImage*, *101*, 404–415.